SIAM DATA MINING 2026 · SALT LAKE CITY

Sculpting the Latent Space

Disentanglement with WAE-MMD and Programmable Priors

Quentin Fruytier[1] · Akshay Malhotra[2] · Shahab Hamidi-Rad[2] · Aditya Sant[2] · Aryan Mokhtari[1] · Sujay Sanghavi[1]

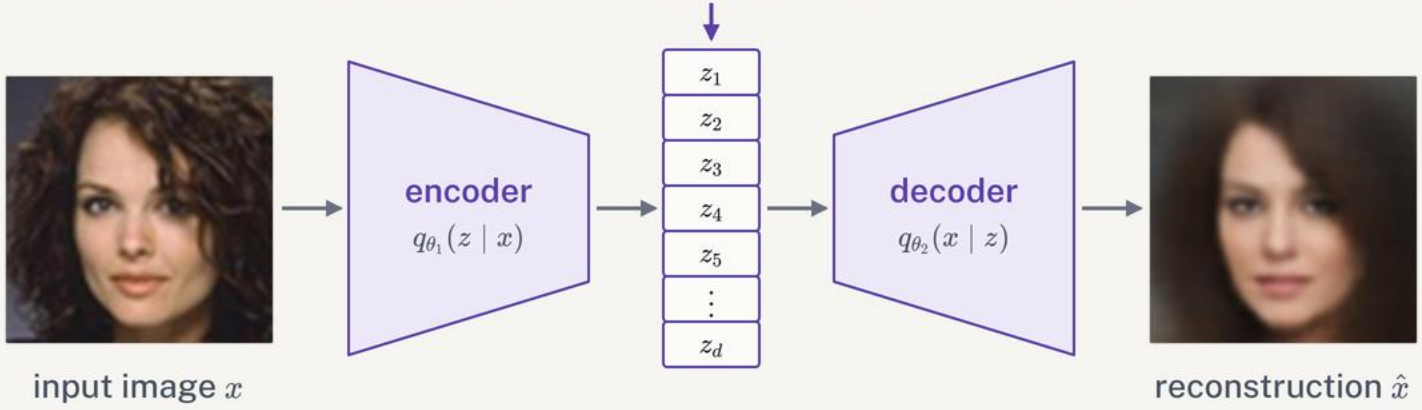
[1] The University of Texas at Austin · [2] InterDigital Communications, AI Lab

THE SETTING

An autoencoder compresses each image into a few numbers

the latent vector $z = (z_1, \ldots, z_d)$: d numbers per image

(d is a design choice; CelebA itself labels 40 attributes)



input image x

reconstruction $\hat{x}$

trained end to end so that $\hat{x} \approx x$

Encode every image: one dot each



$$q_{\theta_1}(z) = \mathbb{E}_{x \sim p(x)}\left[q_{\theta_1}(z \mid x)\right]$$

the aggregate posterior: how the codes of all images are spread

CelebA test image · reconstruction and codes from a plain autoencoder (d = 32) retrained for this talk

THE GOAL · DISENTANGLEMENT

Each latent should be independent and mean one thing

$$z = [\, z_1 \mid z_2 \mid z_3 \mid z_4 \mid \cdots \mid z_{32} \,]$$

	smiling	eyeglasses	blond hair	male
CelebA images labelled no · yes				

(1) Independent $p(z) = \prod_i p(z_i)$

Over all images, one latent says nothing about another.

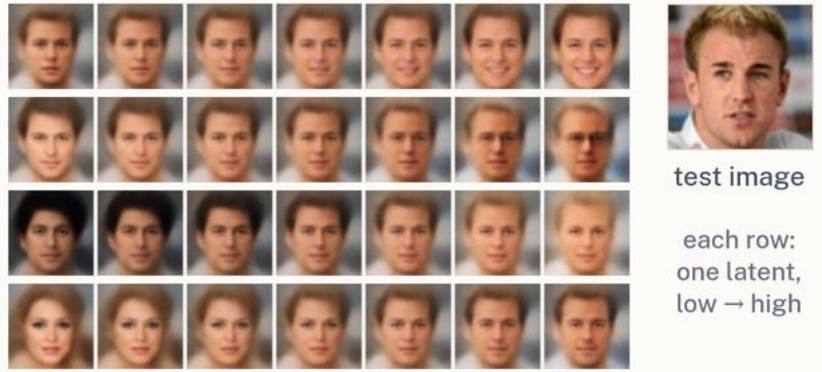


plain autoencoder
its most dependent pair: r = 0.63

ours (SWAE-MMD)
its most dependent pair: r = 0.00

(2) Interpretable

Our model, 4 labelled attributes: move one latent, one thing changes.



test image

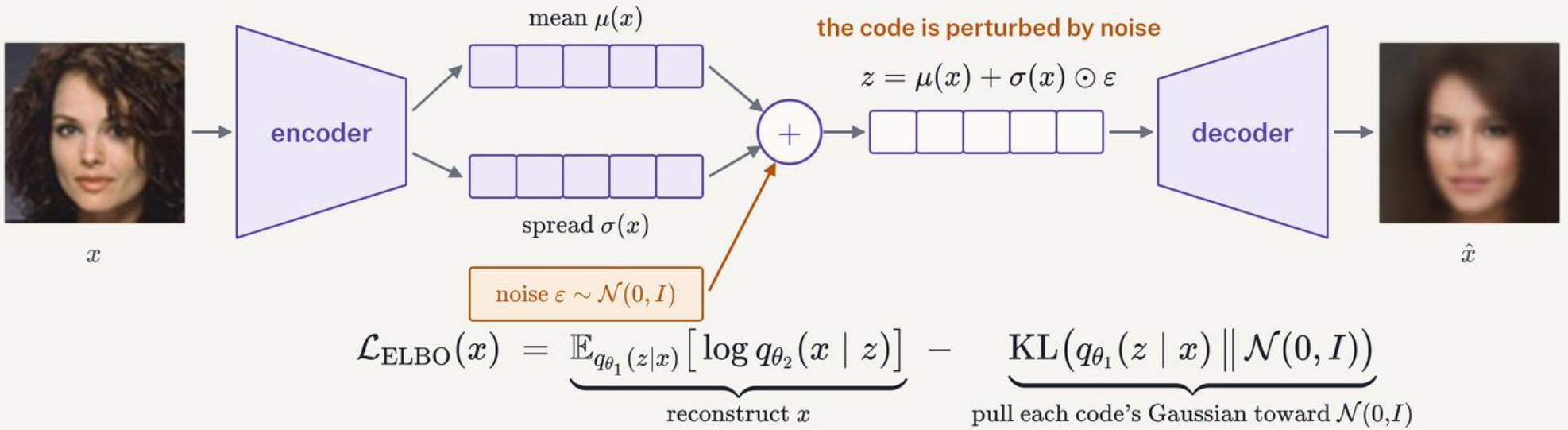
each row: one latent, low → high

The catch: without labels, (2) is not identifiable. So we enforce (1), then add (2) through the prior.

CelebA test images · models retrained with the public code, d = 32 · (2): 4 labelled attributes

THE STANDARD RECIPE

The variational autoencoder adds noise and a KL penalty



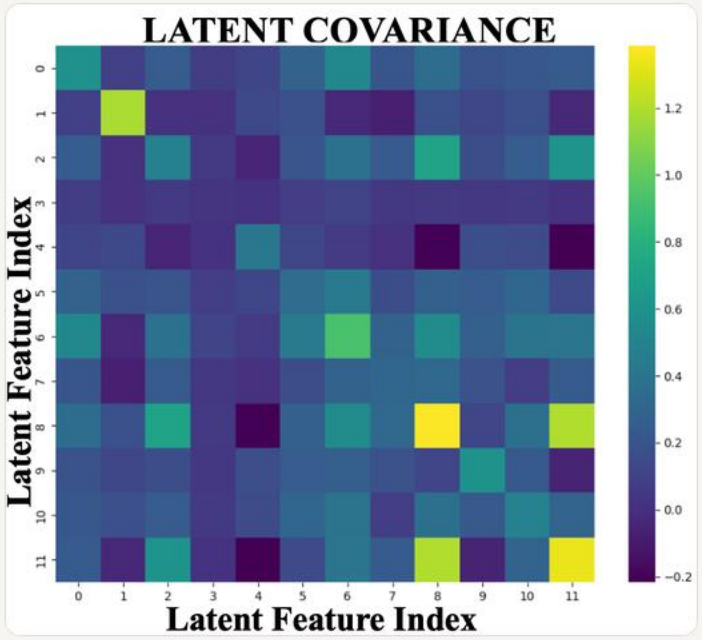
What we need: the **aggregate** q(z) = E_x[q(z | x)] to match the factorized N(0, I).

The VAE hopes to get there by pulling every q(z | x) toward N(0, I), one input at a time.

Kingma & Welling 2014 · reconstruction: a β-VAE retrained on CelebA for this talk (same test face)

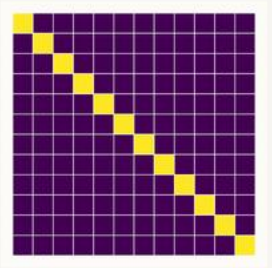
OUR MEASUREMENT · STANDARD VAE ON MNIST, D = 12

Over the dataset, the codes are neither decorrelated nor Gaussian

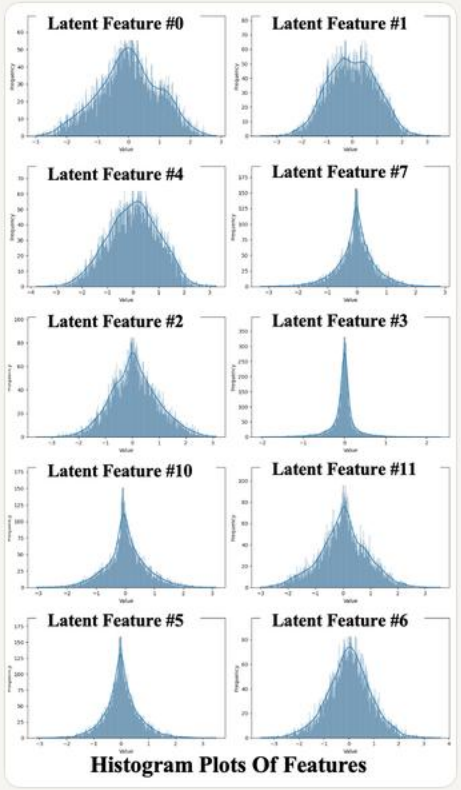


covariance of the 12 latents over all of MNIST

N(0, I) asks for:



the identity: no correlation



marginal of each latent

N(0, I) asks for:



a bell curve in every latent

If every q(z | x) were N(0, I), the aggregate would be too. Here it is not: correlated latents, peaked and skewed marginals.

Paper, Figure 1(a): standard VAE trained on MNIST with 12 latents

WHY IT FAILS

The KL term pulls each code toward 0, one input at a time

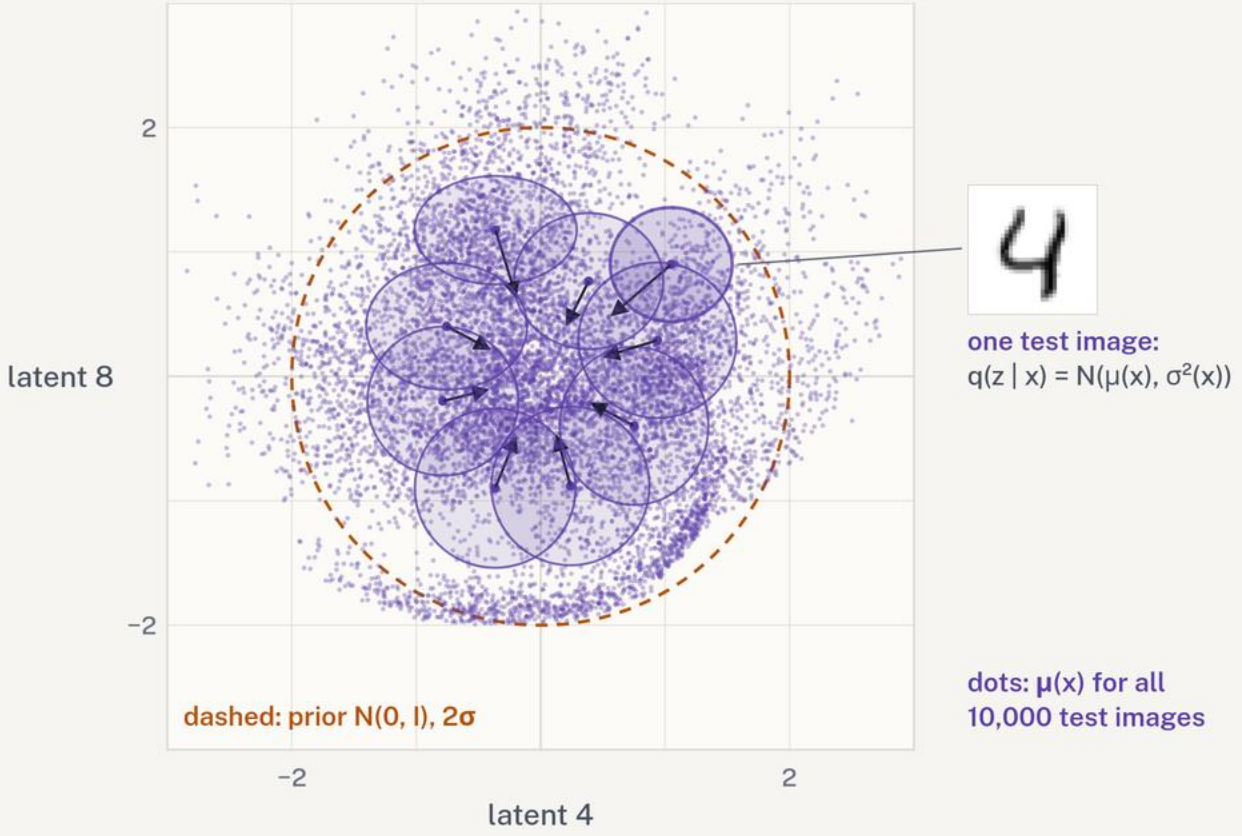
$$\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0,1)\big) = \tfrac{1}{2}\big(\mu^2 + \sigma^2 - 1 - \log \sigma^2\big)$$

For each input and each latent: pull the mean μ to 0 and the spread σ to 1.

Every input is pulled on its own, straight toward 0.
No term compares inputs.

Taken all the way, the pull switches latents off:
6 of 12 latents sit at μ = 0, σ = 1 for every image.

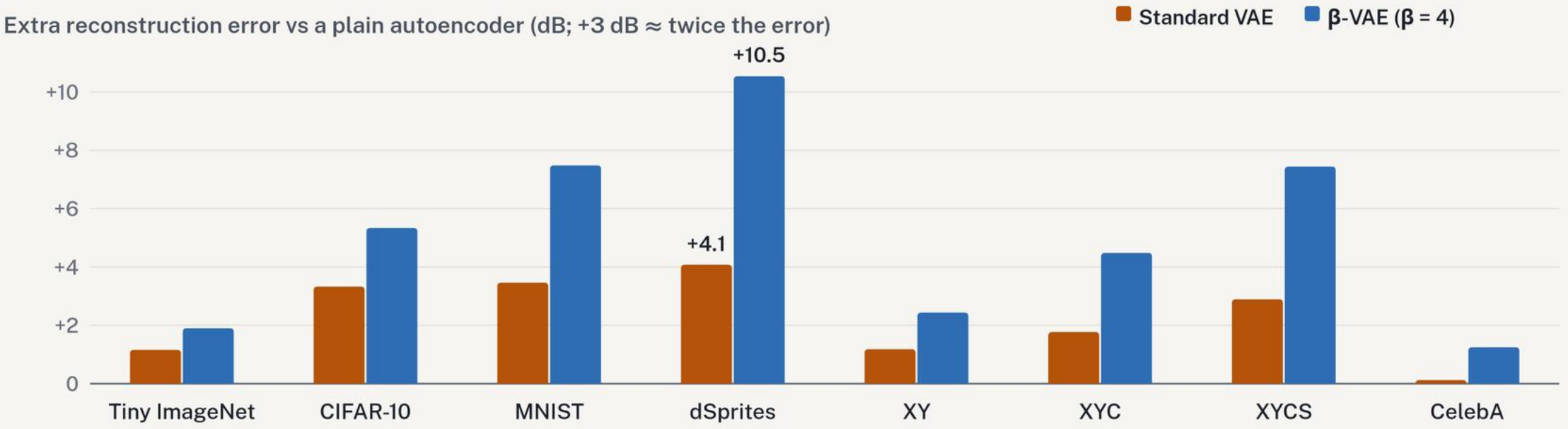
The rest form a curved, clumped cloud that the KL never looks at.



β-VAE (β = 4) on MNIST, d = 12, retrained with the public code · ellipses: 2σ of q(z | x); arrows: the pull on μ (∂KL/∂μ = μ)

THE OTHER COST · RECONSTRUCTION

VAE noise trades reconstruction for generation



Noise and the pull to N(0, I) buy generation and cost reconstruction: +2.2 dB (VAE), +5.1 dB (β-VAE) on average. Compression and editing need a deterministic autoencoder, with no widely accepted way to disentangle one.

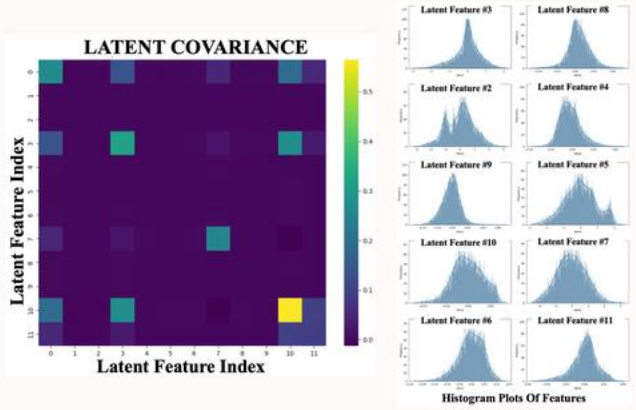
Reconstruction NMSE from the paper's Table 1, minus the standard AE's on the same dataset

TWO COMMON FIXES, AND OURS · MNIST, D = 12

Only matching the whole aggregate distribution works

β-VAE

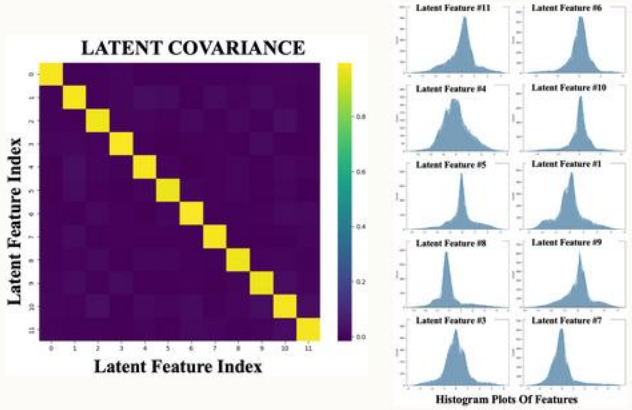
a stronger KL on each code (β = 4)



Still correlated; marginals still not Gaussian. **LPS 0.90**

AE + batch KL

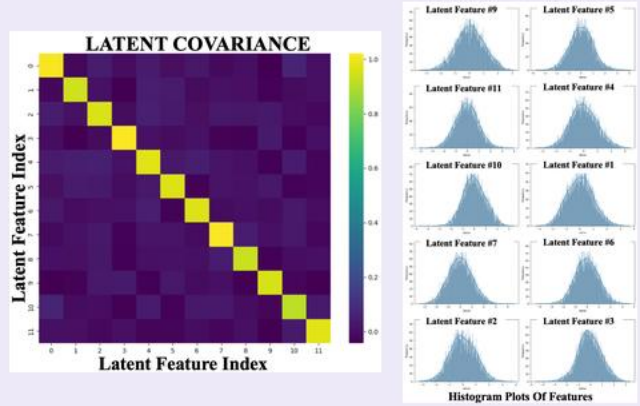
KL on the batch's mean and covariance



Covariance now diagonal, marginals far from Gaussian: still dependent. **LPS 0.97**

SWAE-MMD (ours)

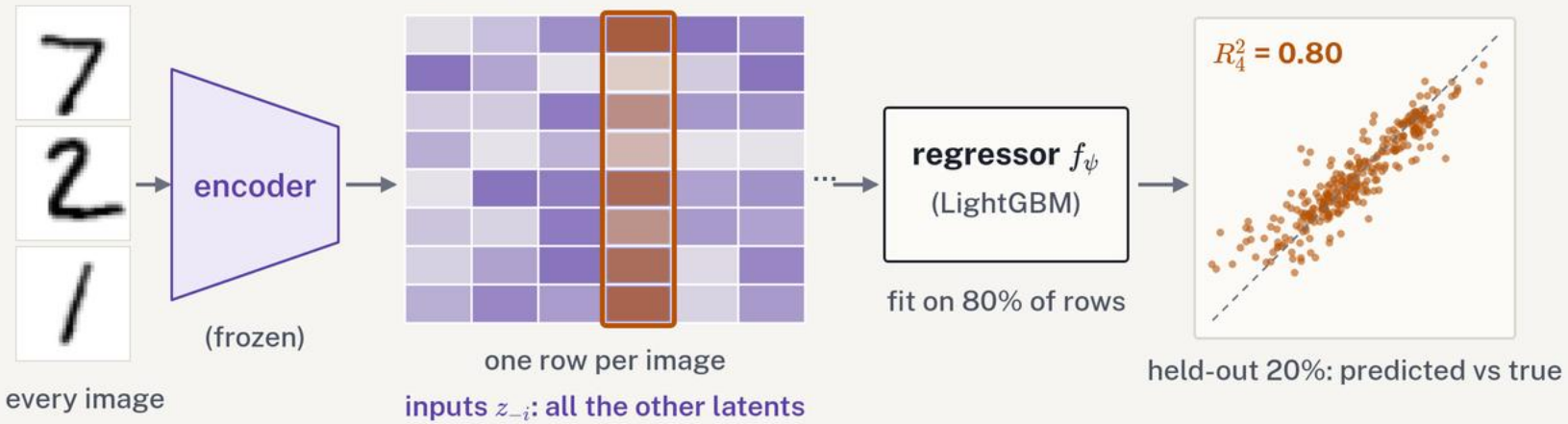
MMD between the batch and prior samples



Diagonal covariance and Gaussian marginals. **LPS 0.22**

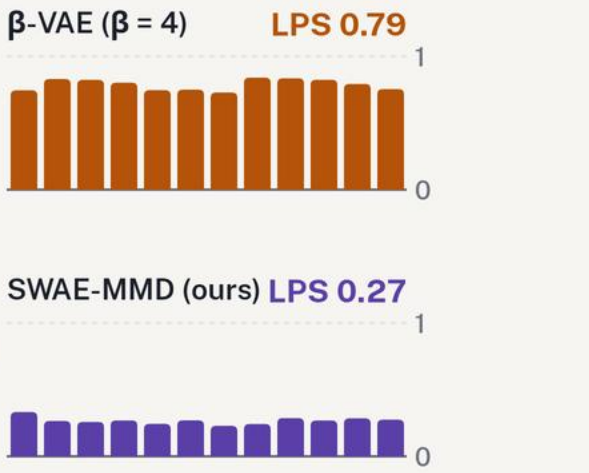
Left: latent covariance over the dataset · right: the latents' marginals · LPS (lower = more independent) is defined next

Can one latent be predicted from the others?



repeat for every latent, average:

$$\mathrm{LPS} = \frac{1}{d}\sum_{i=1}^{d} R_i^2$$



MNIST, d = 12, both retrained (paper: 0.90 and 0.22)

LightGBM, 80/20 split · MNIST, d = 12, retrained with the public code · LPS 0: independent, 1: fully predictable

WHY THE SCORE IS SAFE TO TRUST

LPS can only underestimate entanglement

R^2 of any regressor $\leq \eta^2$, the share of a latent's variance the others truly explain

No regressor beats the conditional mean, so an LPS of 0.5 proves at least half of the latent information is entangled. No labels needed.

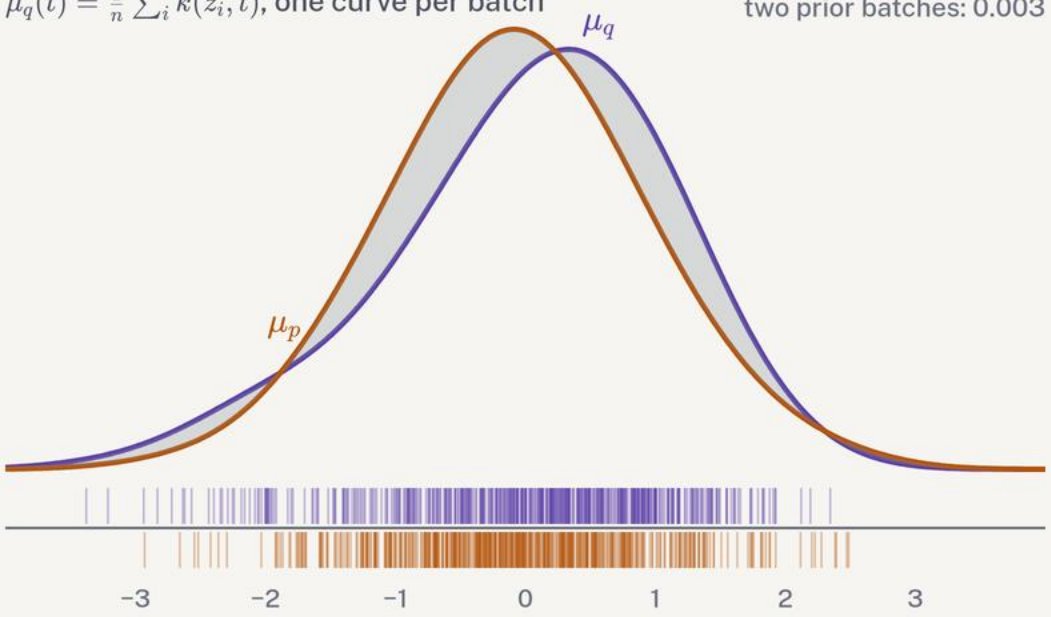
THE TOOL · MAXIMUM MEAN DISCREPANCY

Comparing two distributions from samples alone

q(z) has no formula, only samples (a batch of codes). The prior we can sample too.

average a Gaussian bump over each batch:
$\mu_q(t) = \frac{1}{n}\sum_i k(z_i, t)$, one curve per batch

MMD² = 0.021
two prior batches: 0.003



above: 512 codes, latent 8 of a plain autoencoder on MNIST, standardized
below: 512 samples from the prior N(0, 1)

Why one curve can tell distributions apart

With the kernel $k(x, y) = e^{xy}$, each point becomes an infinite vector

$$\varphi(z) = \left(1,\ z,\ \frac{z^2}{\sqrt{2!}},\ \frac{z^3}{\sqrt{3!}},\ \dots\right)$$

Averaged over a batch it lists every moment; read at t, it is the MGF:

$$\mu_q = \left(1,\ \mathbb{E}z,\ \frac{\mathbb{E}z^2}{\sqrt{2!}},\ \frac{\mathbb{E}z^3}{\sqrt{3!}}, \dots\right), \qquad \langle \mu_q, \varphi(t)\rangle = \mathbb{E}\, e^{tz}$$

	$\mathbb{E}z$	$\mathbb{E}z^2$	$\mathbb{E}z^3$	$\mathbb{E}z^4$
codes	0.00	1.00	**-0.54**	3.13
prior	0	1	**0**	3

Mean and variance agree, so a KL on batch statistics is satisfied; the third moment does not, and MMD sees it.

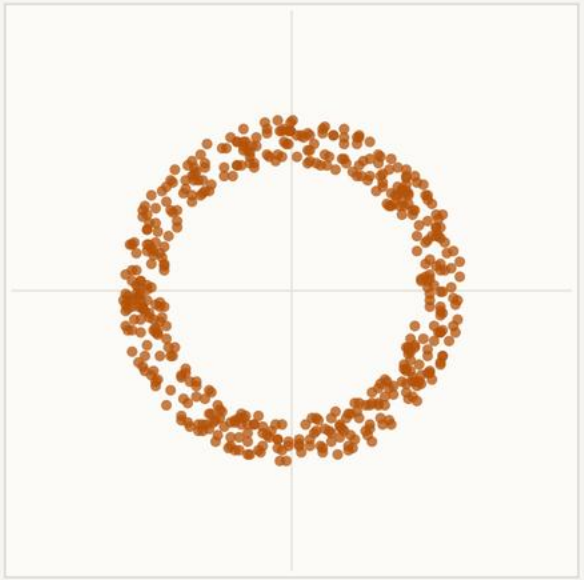
Our Gaussian kernel holds the same moments, damped by $e^{-z^2/2\sigma^2}$ so they always exist: MMD = 0 only if $q = p$.

Gretton et al., JMLR 2012 · codes: a plain autoencoder on MNIST (d = 12) retrained for this talk · Gaussian kernel, σ = 0.5

SCULPTING, ONE BATCH AT A TIME

MMD pulls the codes onto any prior

loss = reconstruction error + λ · MMD²(codes, prior samples)

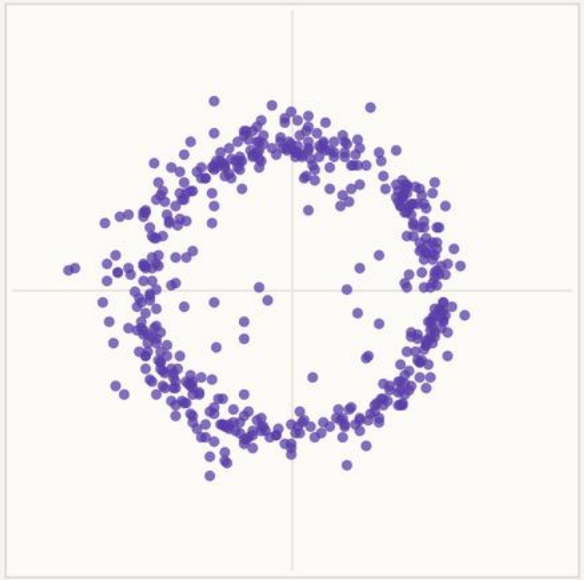


prior p(z): a ring
samples we draw

training step
9,360

MMD²(codes, prior)
0.007

second run: only the prior changed



codes z = encoder(x)
500 MNIST test images, 2 latents

Two real runs: MNIST, 2 latents, public code, λ = 20 · the same 500 test images at logged steps

OUR METHOD · SWAE-MMD

WAE-MMD, extended to a programmable prior

WAE-MMD · Tolstikhin et al., ICLR 2018

$$\mathcal{L}_{\text{WAE}} = \mathbb{E}_{p(x)}\big[\log q_\theta(x)\big] - \lambda\,\text{MMD}^2\big(q_{\theta_1}(z),\, p(z)\big)$$

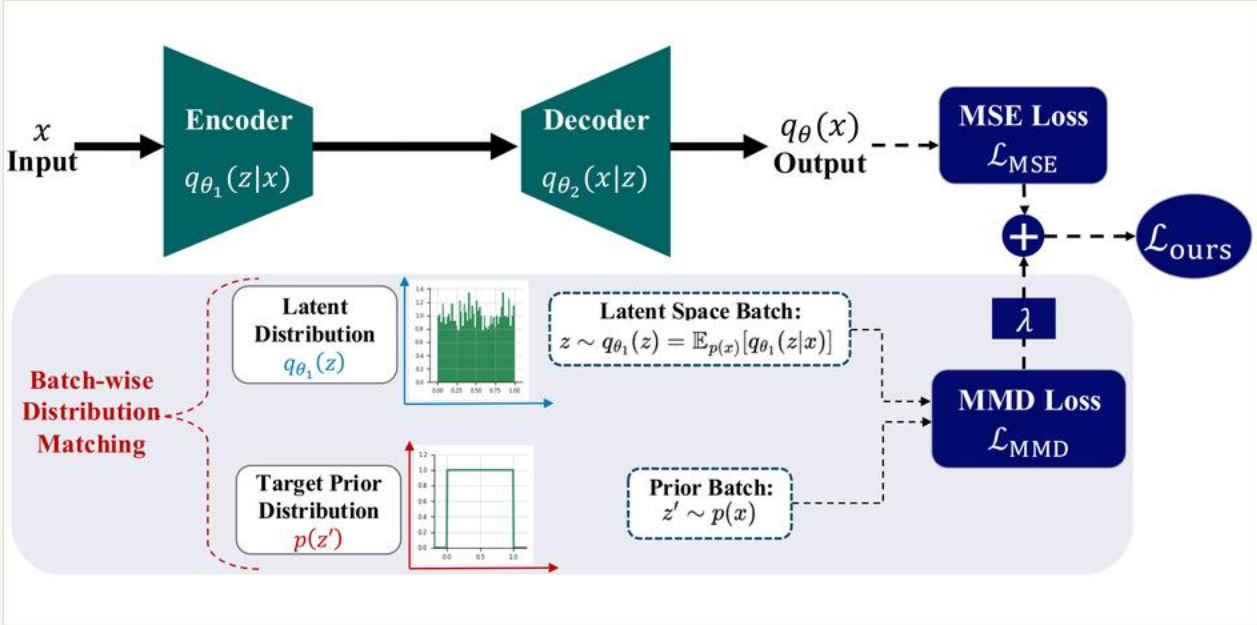
$p(z) = \mathcal{N}(0, I)$, built to generate images

SWAE-MMD · ours

$$\mathcal{L}_{\text{ours}} = \mathbb{E}_{p(x)}\big[\log q_\theta(x)\big] - \lambda\,\text{MMD}^2\big(q_{\theta_1}(z, f_\phi(u)),\, p(z, f_\phi(u))\big)$$

- any prior p we can sample: factorized → independence; its shape → compression
- joint with pseudo-labels $f_\phi(u)$ → alignment to chosen factors

No labels and $p = \mathcal{N}(0, I)$: exactly WAE-MMD. Goal: representations, not generation.

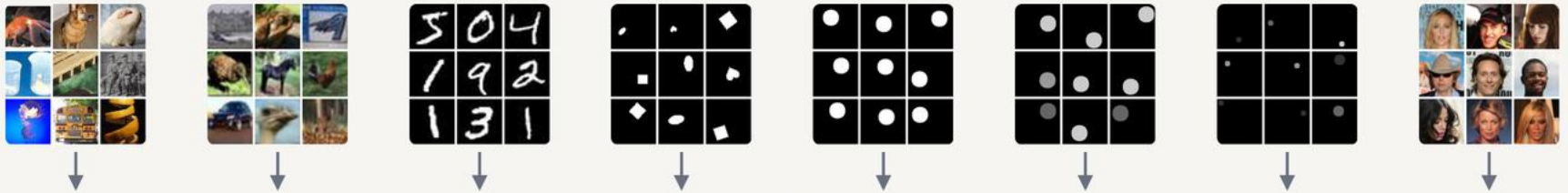


The MMD sees a whole batch of codes, so it shapes the aggregate.

RESULTS · 8 DATASETS

70% less entanglement than the next best method

a sample of each dataset



LPS ↓ per model and dataset: how well the other latents predict each latent (mean R^2). 0 = independent, 1 = fully predictable.

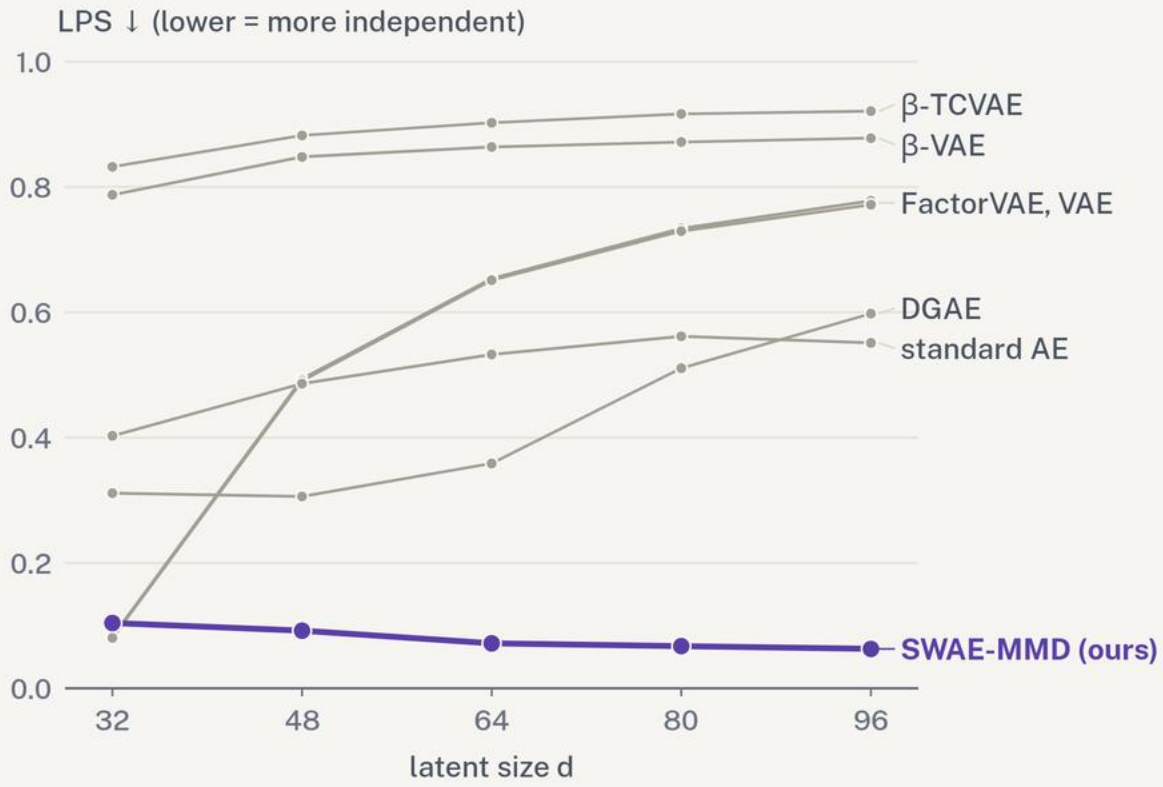
model · LPS ↓	ImageNet [64]	CIFAR-10 [64]	MNIST [12]	dSprites [5]	XY [2]	XYC [3]	XYCS [4]	CelebA [32]
SWAE-MMD (ours)	**0.03**	**0.04**	**0.22**	**0.04**	−0.05	**−0.01**	**0.11**	0.08
FactorVAE (γ = 10)	0.44	0.63	0.27	0.08	−0.04	0.00	0.17	**0.07**
NashAE	0.56	0.17	**0.22**	0.10	**−0.07**	0.14	0.37	0.36
β-TCVAE (β = 4)	0.79	0.89	0.84	0.22	−0.04	**−0.01**	0.17	0.59
β-VAE (β = 4)	0.73	0.85	0.90	0.52	−0.05	0.00	0.28	0.53
DGAE	0.77	0.69	0.63	0.56	0.25	0.50	0.59	0.48

Averaged over the 8 datasets, LPS falls by 70% against FactorVAE, with reconstruction as good or better (CIFAR-10: −15.61 dB NMSE vs −12.49 dB).

Table 1 of the paper · latent size in brackets · ImageNet = Tiny ImageNet · bold = best in column

RESULTS · CIFAR-10

Independence holds as the latent space grows



Flat in the latent size

From 32 to 96 latents ours stays at 0.10 or below; the baselines grow more entangled.

Insensitive to its knobs

Batch 64–2048, λ 0.1–5, 1–5 kernel widths: small changes only.

CIFAR-10 · values from the paper's appendix figure (LPS with LightGBM), redrawn; its error bars are smaller than the markers

PROGRAMMABLE PRIOR · COMPRESSION

A quantizer only works if the codes have the shape it expects

A b-bit quantizer snaps each latent to one of 2^b levels

uniform, built for a uniform latent (2 bits)



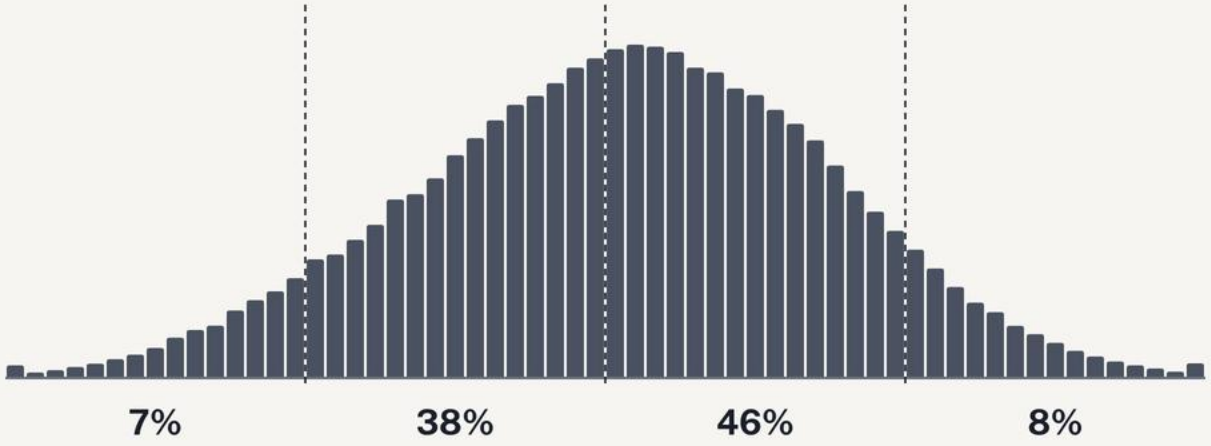
Lloyd–Max, built for N(0, 1) (2 bits)



dots: levels · dashed: bin edges · grey: the density it is built for

Real codes, same uniform 2-bit quantizer

plain autoencoder on MNIST, all 12 latents, each over its own range



share of codes per level · a uniform latent would put 25% in each

So we set the prior to the quantizer's distribution and train with the quantizer in the loop.

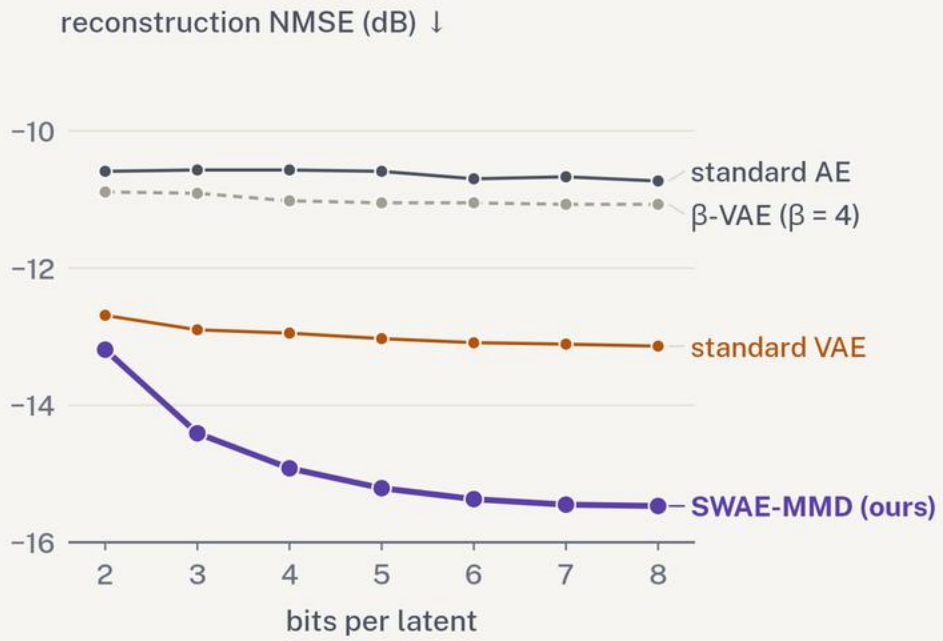
Left: 2-bit designs · right: a plain autoencoder on MNIST (d = 12) retrained for this talk, each latent over its 0.1–99.9% range

COMPRESSION RESULTS · CIFAR-10

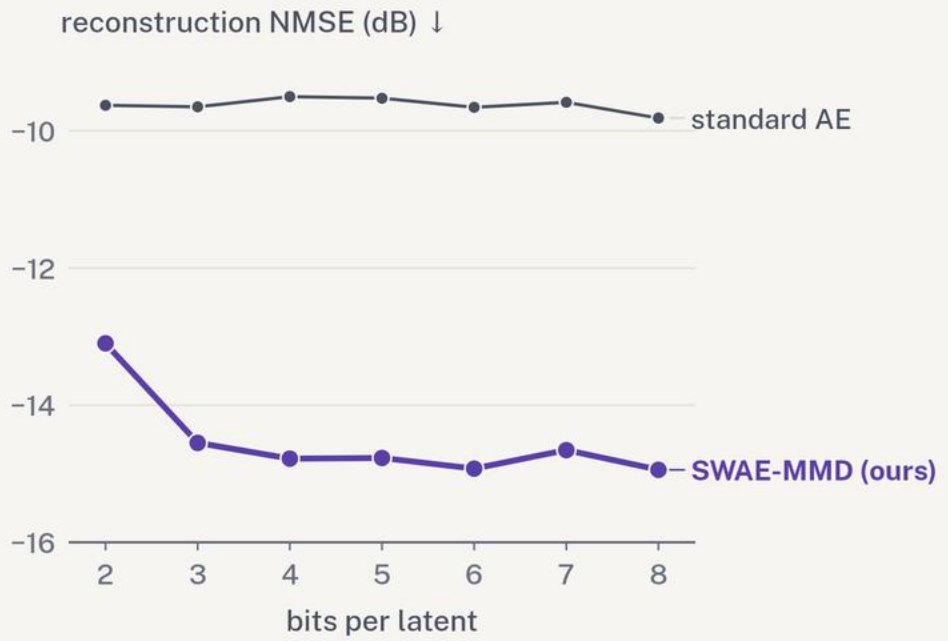
Shape the codes to fit the quantizer

Train with the quantizer in the loop and a prior equal to the distribution the quantizer assumes, Gaussian or uniform.

Gaussian quantizer · our prior: N(0, I)



Uniform quantizer · our prior: uniform



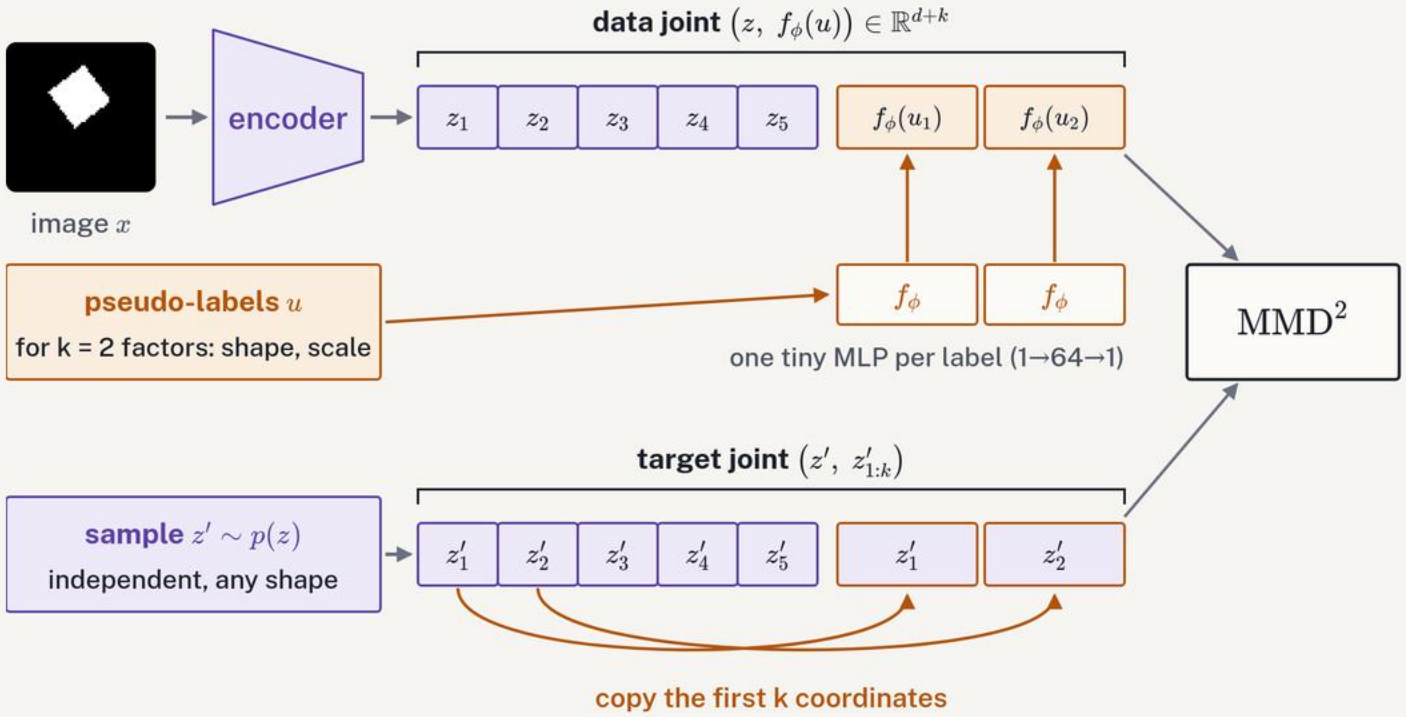
2–7 dB lower error at 2 to 8 bits per latent, with no vector quantization or entropy coding.

CIFAR-10, the paper's quantization figure redrawn · Gaussian-quantizer values read from the published image (±0.05 dB)

PROGRAMMABLE PRIOR · ALIGNMENT

Pin chosen latents to a few labels through the prior

Independent latents are still arbitrary: nothing says which one is shape. Labels for a few factors fix that.



To match the target, the data side must copy too:

$$z_1 = f_\phi(u_1), \quad z_2 = f_\phi(u_2)$$

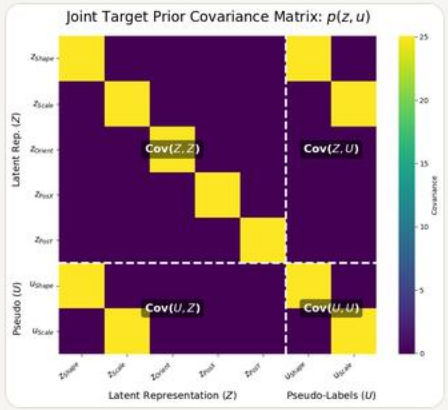
latent 1 now carries shape, latent 2 carries scale, and z still follows $p(z)$: independent and shaped.

f_ϕ only rescales each label onto the prior's marginal. Images without labels use $(z,\ z_{1:k})$ and satisfy the copy for free.

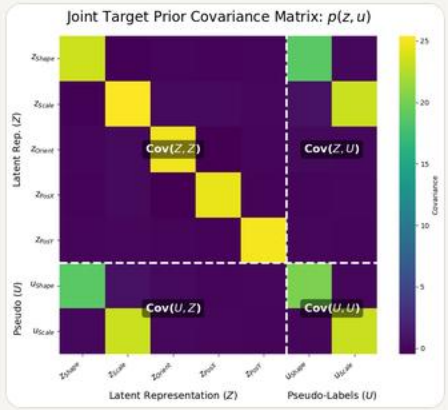
dSprites: 5 factors (shape, scale, orientation, x, y); k = 2 labelled here · label maps: 1→64→1 MLPs with BatchNorm

ALIGNMENT RESULTS · DSPRITES

A few pseudo-labels lock latents to real factors

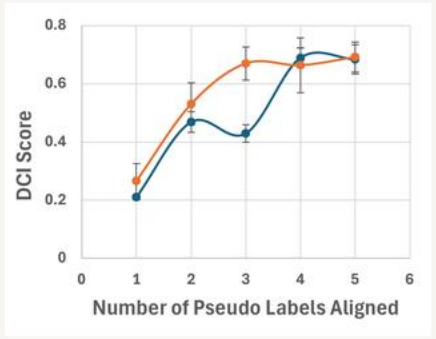


target

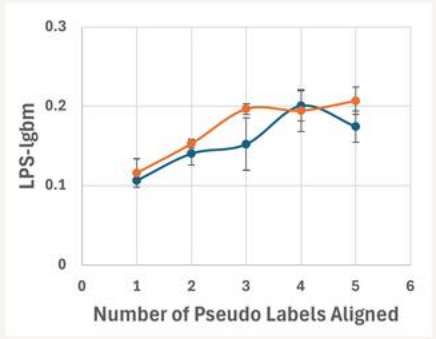


learned

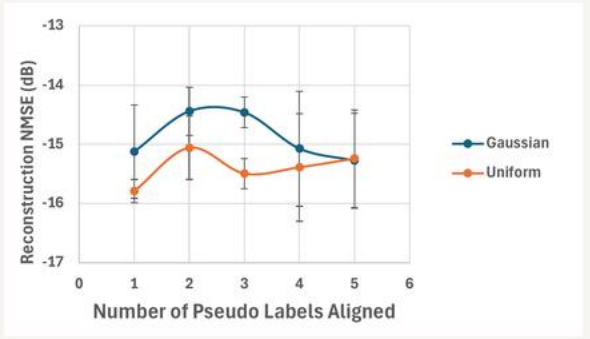
Learned matches target: the two labels sit on latents 1 and 2, and the latents stay independent.



DCI disentanglement ↑



LPS ↓



reconstruction NMSE (dB) ↓

x-axes: pseudo-labels aligned (of 5) · blue: Gaussian prior, orange: uniform prior · DCI always against all 5 factors

TAKEAWAYS

The prior is an engineering tool, not a fixed assumption

A label-free test

LPS lower-bounds entanglement and exposes the VAE-style failure.

A regularizer that works

MMD on the aggregate posterior: 70% less entanglement, reconstruction kept.

One loss, many priors

Semantic alignment, quantizer matching and latent copying.

JOINT WORK WITH



Akshay Malhotra
InterDigital



Shahab Hamidi-Rad
InterDigital



Aditya Sant
InterDigital



Aryan Mokhtari
UT Austin



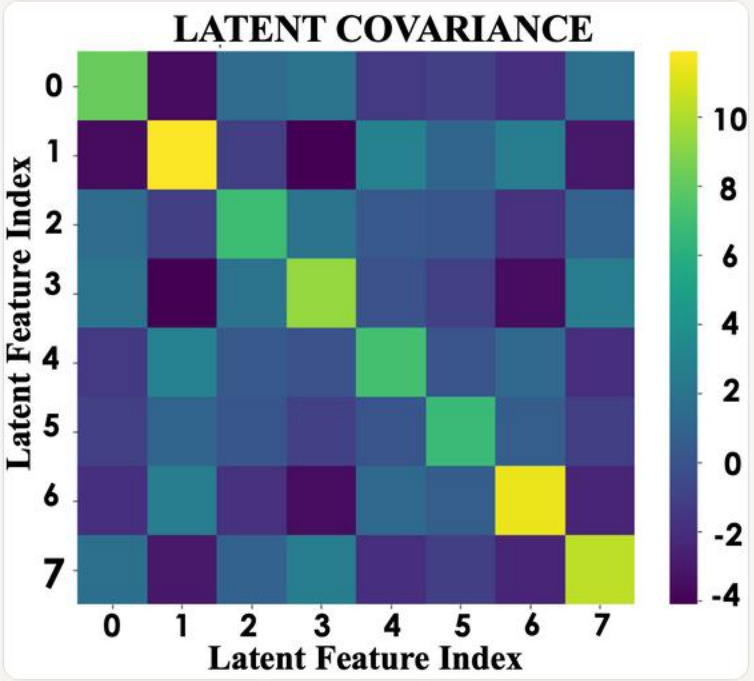
Sujay Sanghavi
UT Austin

arxiv.org/abs/2510.11953
github.com/Quentin-Fruytier/Sculpting-Latent-Spaces-With-MMD
Work done at InterDigital AI Lab with the University of Texas at Austin · thank you

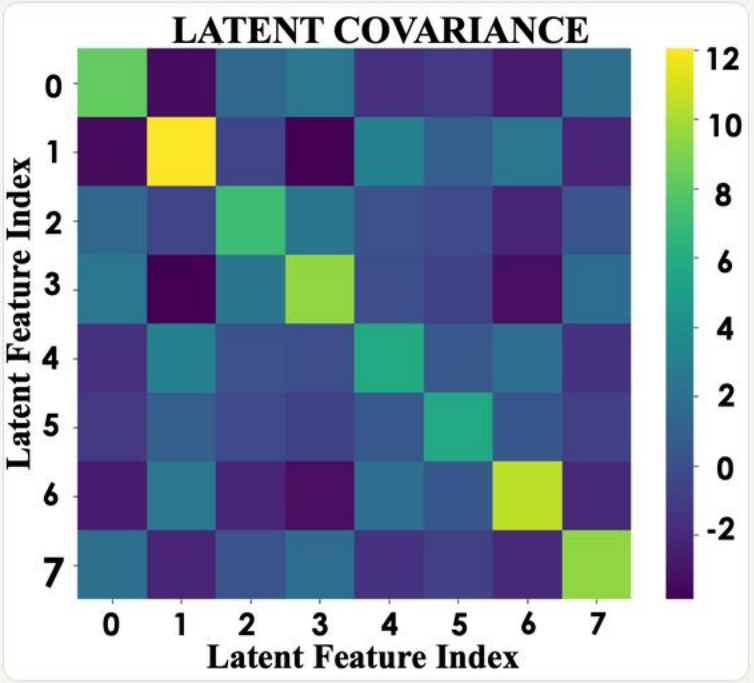
BACKUP · A PRIOR IS ANYTHING YOU CAN SAMPLE

MMD can copy another model's latent space

teacher: a standard AE's codes, used as the prior



student: trained with MMD to match them



An entangled, co-dependent geometry that no closed-form KL target could express.

Covariance and marginals come out nearly identical.

MNIST, latent covariance over the dataset · paper appendix, latent copying experiment

BACKUP · ALIGNMENT SCORE (DCI)

Without labels we optimize independence, not alignment

	dSprites		XY		XYC		XYCS	
model	DCI ↑	LPS ↓	DCI ↑	LPS ↓	DCI ↑	LPS ↓	DCI ↑	LPS ↓
SWAE-MMD, Gaussian prior (ours)	0.025	**0.039**	0.199	−0.053	0.113	**−0.014**	0.048	**0.111**
SWAE-MMD, hybrid prior (ours)	0.196	0.105	0.469	**−0.065**	**0.227**	0.168	0.146	0.203
DGAE	0.219	0.557	**0.521**	0.248	0.175	0.498	0.187	0.596
FactorVAE	0.079	0.082	0.336	−0.040	0.076	−0.000	0.148	0.187
β-TCVAE	0.195	0.222	0.368	−0.042	0.090	−0.010	0.245	0.171
β-VAE	0.217	0.525	0.358	−0.053	0.138	0.003	**0.262**	0.276
standard AE	0.041	0.187	0.362	−0.028	0.085	0.042	0.068	0.265
standard VAE	**0.225**	0.100	0.102	−0.055	0.085	0.016	0.105	0.180

Unsupervised, our Gaussian model is the most independent but not aligned (DCI 0.03 on dSprites). With pseudo-labels for 3–4 of the 5 factors its DCI reaches about 0.65–0.67 (slide 19).

DCI can reward dependent latents: DGAE has the best DCI on XY (0.52) with LPS 0.25.

Paper appendix (synthetic datasets) · DCI: random-forest probe (Eastwood & Williams 2018) · bold = best in column